



COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 03

**BOOLEAN THEOREMS AND
GATES** -1



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Recap of Previous Lecture

Imp basics & Theorems



Topics to be Covered

Logic gates



[GATES]

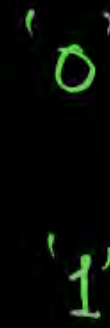


- Basic Gates : NOT gate, OR gate, AND gate
- Arithmetic Gates : XOR gate, XNOR gate
- Universal Gates : NAND gate, NOR gate

Diagram illustrating the implementation of NOT using a NAND gate:

Left side: Input A is connected to both inputs of a NAND gate. The output is $\overline{A} = y$.

Right side: Input A is connected to both inputs of a NAND gate. The output is $\overline{A} = y$.


$$\begin{aligned} \gamma(A) &= \Sigma(0) = \pi(1) \\ &= \overline{A} = \overline{A} \end{aligned}$$

[AND GATE]



Representation :

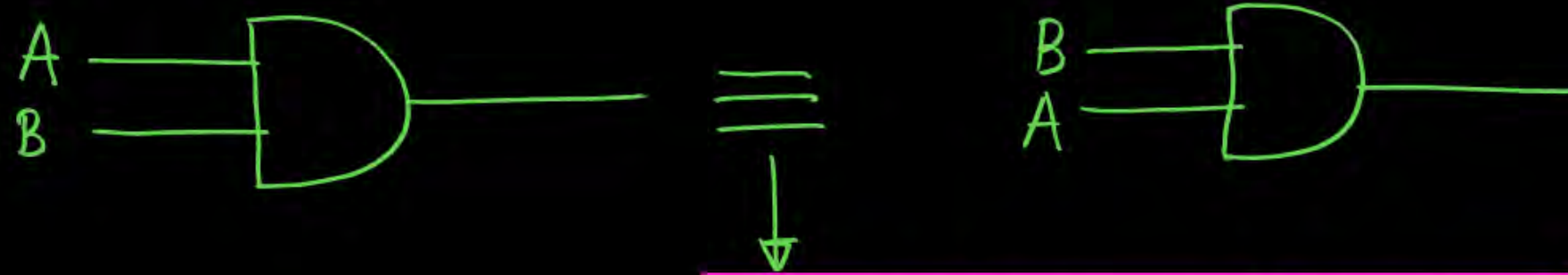


	2^1	2^0	
	A	B	Output $y = A.B$
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

$$y(A,B) = \sum(3) = \pi(0,1,2)$$
$$= A.B$$

$$= (A+B) \cdot (A+\bar{B}) (\bar{A}+B)$$
$$= (A+B\bar{B}) (\bar{A}+B)$$
$$= A(\bar{A}+B)$$
$$= 0 + A.B = A.B$$

- Commutative Law :



It holds commutative law

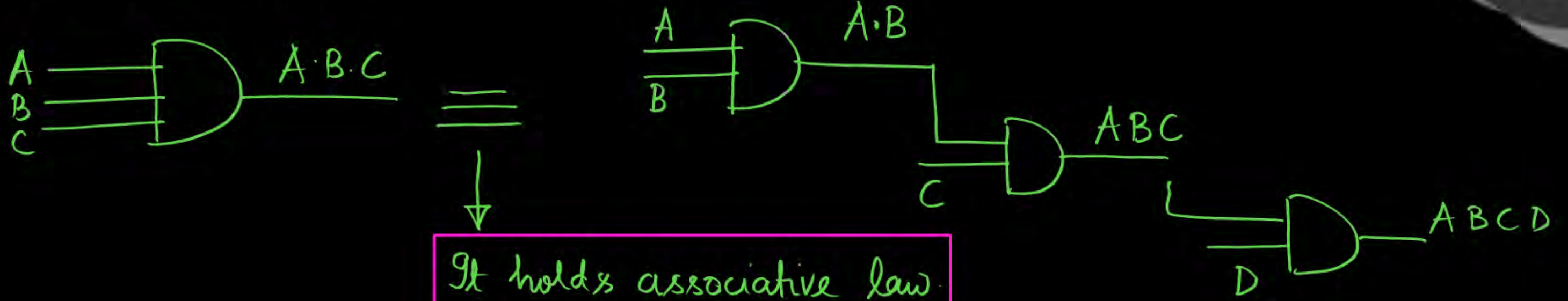


Position of variables is irrelevant.

$$AB = BA$$

$$ABC = ACB = BAC$$

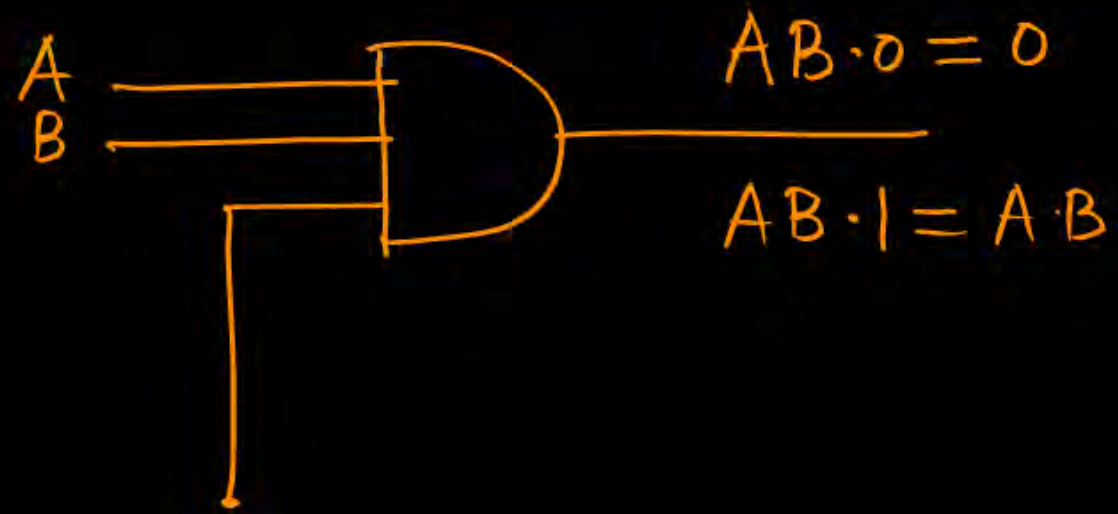
- Associative Law :



It holds associative law.

→ means we can implement multi i/p AND operation using 2-i/p AND gate.

- Enable and Disable input for AND gate :



Control i/p = 0 [0 - disable i/p for AND gate]
 = 1 [1 → enable i/p for AND gate]

- IMP points :

- If any one of the i/p line is at logic '0' then irrespective of other i/p lines o/p has to be '0'.
- O/P will be '1' only if all the i/Ps are at logic '1' together.

[OR GATE]



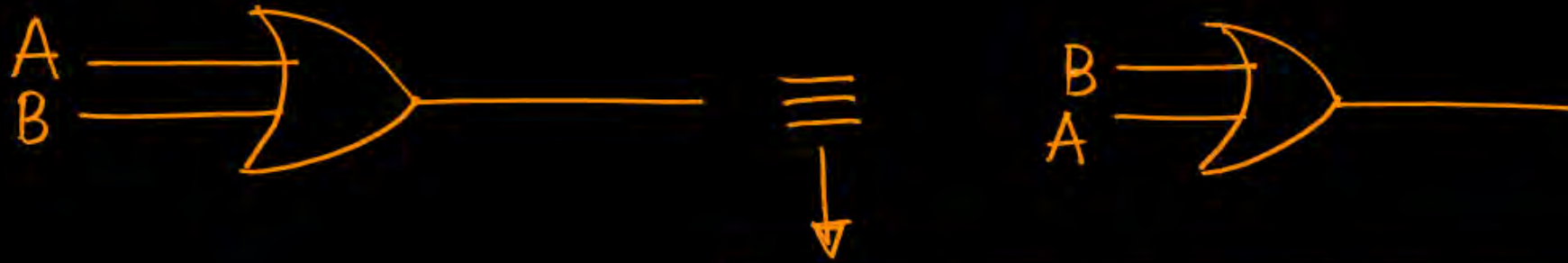
Representation :



	A	B	$y = A + B$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

$$\begin{aligned} y(A,B) &= \sum (1, 2, 3) = \pi(0) \\ &= \bar{A}B + A\bar{B} + AB = (A+B) \\ &= A + \bar{A}B \\ &= (A + \bar{A}) \cdot (A + B) \\ &= (A + B) \end{aligned}$$

- Commutative Law :



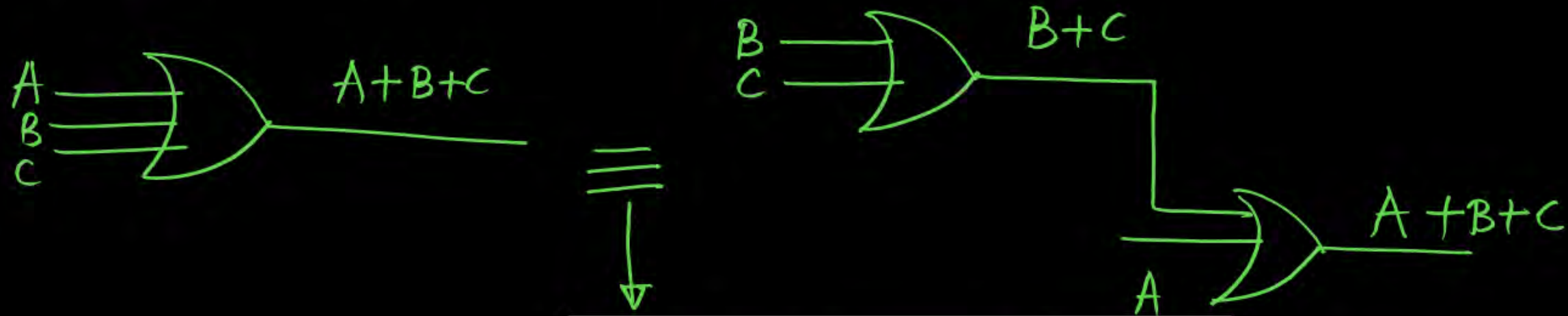
It holds Commutative law

↓
position of variables is irrelevant.

$$A + B = B + A$$

$$\begin{aligned} A + B + C &= B + A + C \\ &= C + A + B \\ &= A + C + B \end{aligned}$$

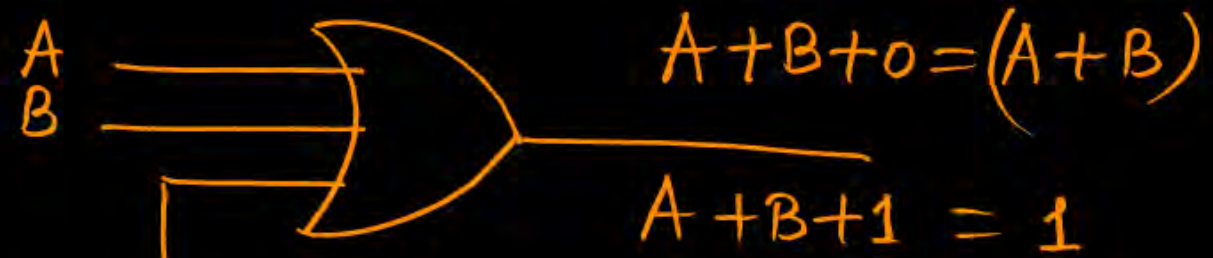
- Associative Law :



It holds associative law.

→ means we can implement multi i/p OR operation using 2-i/p OR operation.

- Enable and disable input for 'OR gate' :



Control i/p = 0 [0 - enable i/p for OR gate]

= 1 [1 → disable i/p for OR gate]

- IMP Points :

- If any one of the i/p line is at logic '1' then irrespective of other i/p lines o/p will be '1'.
- O/p will be '0' only if all the i/p lines together are at logic '0'.

[XOR GATE] (Exclusive OR)

→ it is defined for 2-i/Ps only.

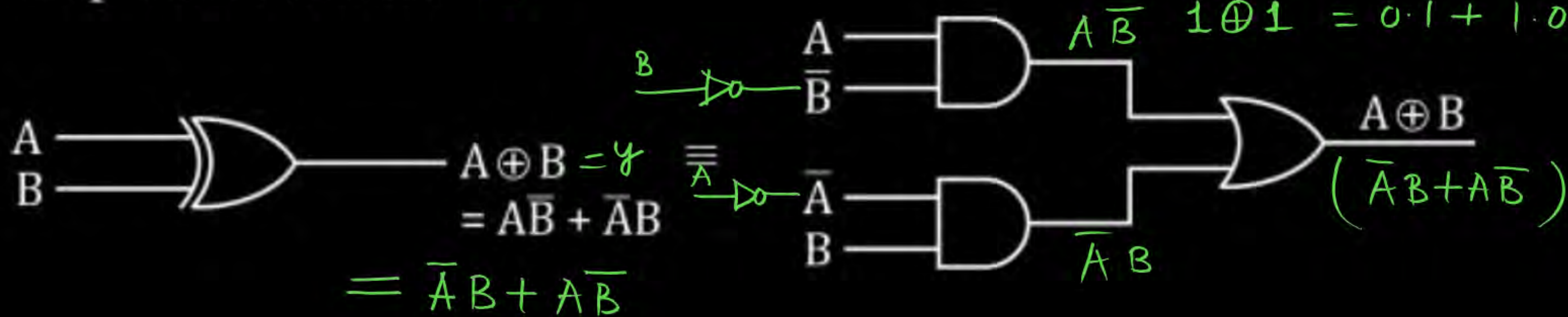
$$0 \oplus 0 = 1 \cdot 0 + 0 \cdot 1 = 0$$

$$0 \oplus 1 = 1 \cdot 1 + 0 \cdot 0 = 1$$

$$1 \oplus 0 = 0 \cdot 0 + 1 \cdot 1 = 1$$

$$1 \oplus 1 = 0 \cdot 1 + 1 \cdot 0 = 0$$

Representation: $\oplus$



	A	B	$y = A \oplus B$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

$$y(A,B) = \sum(1,2) = \pi(0,3)$$

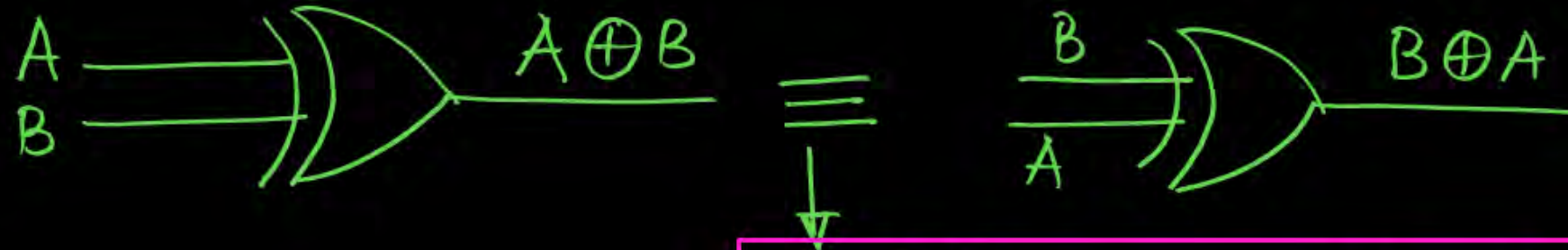
$$= \overline{A}B + A\overline{B} = (A+B) \cdot (\overline{A} + \overline{B})$$

$$\underline{AB} \oplus \underline{BC} = \overline{AB} \cdot BC + AB \cdot \overline{BC}$$

$$\underline{\overline{A}} \oplus \underline{B} = \overline{\overline{A}} \cdot B + \overline{A} \cdot \overline{B}$$

$$\underline{(A+B)} \oplus \underline{B\overline{C}} = \overline{A+B} \cdot B\overline{C} + (A+B) \cdot \overline{B\overline{C}}$$

- Commutative Law :



$$A \oplus B = B \oplus A$$

It holds commutative law.

position of variables is irrelevant.

- Properties of XOR GATE :

- $A \oplus A = 0 = \bar{A} \cdot A + A \cdot \bar{A}$

$$\bar{A} \oplus \bar{A} = 0$$

$$AB \oplus AB = 0$$

- $A \oplus \bar{A} = \bar{A} \oplus A = 1 = \bar{A} \cdot \bar{A} + A \cdot \bar{A}$

$$AB \oplus \overline{AB} = 1$$

- $A \oplus 0 = A = \bar{A} \cdot 0 + A \cdot \bar{0}$

$$0 \oplus A = A$$

$$\bar{A} \oplus 0 = \bar{A}$$

$$AB \oplus 0 = AB$$

- $A \oplus 1 = \bar{A} = 1 \oplus A = \bar{A} \cdot 1 + A \cdot \bar{1}$

$$\bar{A} \oplus 1 = \bar{\bar{A}} = A$$

$$AB \oplus 1 = \overline{AB}$$

- Exchange properties of XOR GATE :

If $A \oplus B = C \rightarrow$ given

then $A \oplus C = B \rightarrow$ true

then $B \oplus C = A \rightarrow$ true

- $A \oplus C = A \oplus A \oplus B = 0 \oplus B = B$

- $B \oplus C = B \oplus A \oplus B = 0 \oplus A = A$

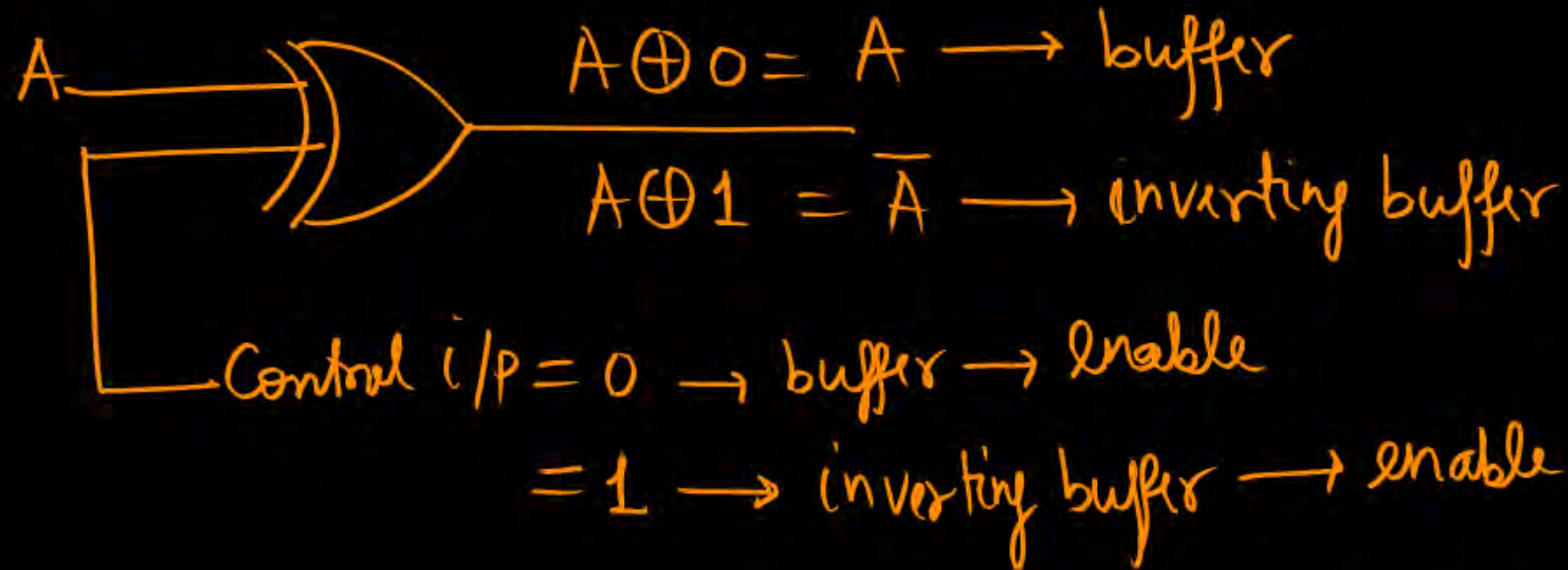


- $A \oplus A \oplus A \oplus A \dots n \text{ times}$ [n - represents no. of A]

then o/p = 0 for n -even

= A for n -odd

- Buffer and inverting buffer using XOR :



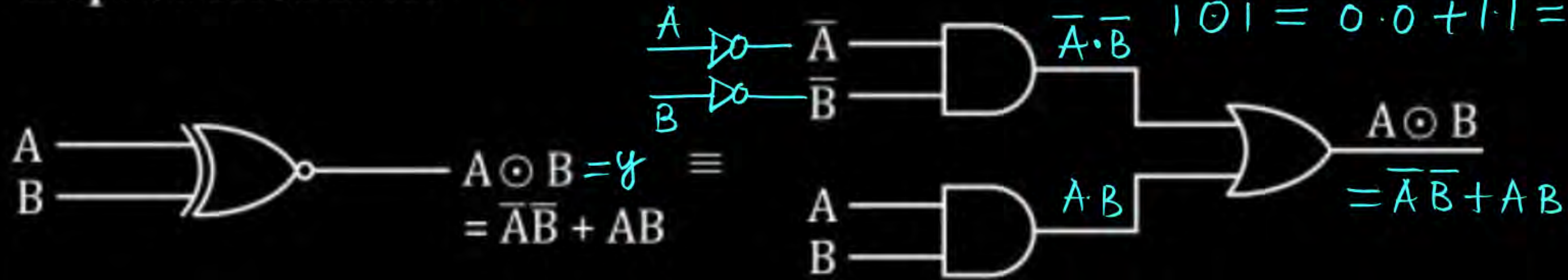
$$A \oplus A = 0$$

$$\underbrace{A \oplus A} \oplus \underbrace{A \oplus A} = 0$$

$$A \oplus A \oplus A = 0 \oplus A = A$$

[XNOR GATE] $\rightarrow$ defined for 2- i/p_s only.

Representation: $\odot$



$$\begin{aligned}
 0 \odot 0 &= 1 \cdot 1 + 0 \cdot 0 = 1 \\
 0 \odot 1 &= 1 \cdot 0 + 0 \cdot 1 = 0 \\
 1 \odot 0 &= 0 \cdot 1 + 1 \cdot 0 = 0 \\
 1 \odot 1 &= 0 \cdot 0 + 1 \cdot 1 = 1
 \end{aligned}$$

	A	B	$y = A \odot B$
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

$$\begin{aligned}
 y(A,B) &= \sum(0,3) = \pi(1,2) \\
 &= \bar{A}\bar{B} + AB \\
 &= (A + \bar{B})(\bar{A} + B)
 \end{aligned}$$

$$\begin{aligned}
 \overline{A \oplus B} &= A \odot B \\
 \overline{A \odot B} &= A \oplus B
 \end{aligned}$$

- $\underline{AB} \odot \underline{BC} = \overline{AB} \cdot \overline{BC} + AB \cdot BC$

$$(A+B) \odot \overline{BC} = \overline{A+B} \cdot \overline{BC} + (A+B) \cdot \overline{BC}$$

$$\overline{A} \odot B = \overline{A} \cdot \overline{B} + \overline{A} \cdot B$$



2 Minute Summary

NOT, AND, OR, XOR, XNOR.

Thank you

GW
Soldiers !

